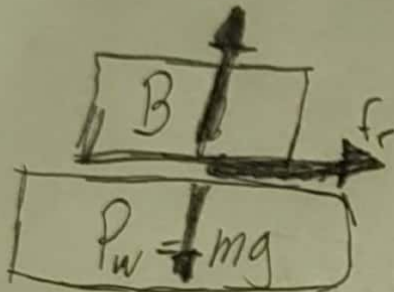


Third Exam Partial

1)



$$\Sigma F = ma$$

$$f_r = ma$$

$$\mu(mg) = ma$$

$$\mu(mg) = mA(4\pi^2 f^2)$$

$$a_{\max} = A\omega^2 = A(4\pi^2 f^2)$$

$$\omega = 2\pi f$$

$$\omega^2 = (2\pi f)^2$$

$$\omega = 4\pi^2 f^2$$

$$A = \frac{\mu g}{4\pi^2 f^2} = \frac{0.600 \times 9.8}{4\pi^2 (1.5)^2} = 0.06619 \approx \underline{0.0662 \text{ m}}$$

$$2) \quad m_1 = 42.5 \text{ kg} \quad T_1 = 1.3 \text{ s}$$

$$m_2 = ? \quad T_2 = 2.54 \text{ s}$$

$$T_1 = 2\pi \sqrt{\frac{m_1}{K}}$$

$$K = \frac{4\pi^2 m_1}{T_1^2} = \frac{4\pi^2 (42.5)}{1.3^2}$$

$$K = 992.8 \text{ N/m}$$

$$T_2 = 2\pi \sqrt{\frac{m_2}{K}}$$

$$m_2 = \frac{K T_2^2}{4\pi^2} = \frac{(992.8)(2.54)^2}{4\pi^2}$$

$$m_2 = 162.24 \text{ kg}$$

$$m_2 = m_{\text{ase}} + m_{\text{gilla}}$$

$$m_{\text{ase}} = m_2 - m_{\text{gilla}}$$

$$= 162.24 - 42.5 = 119.74 \text{ kg}$$

3)

$$m = 8.94 \times 10^{22} \text{ Kg}$$

$$r = 1875 \text{ Km}$$

$$m_i = m_i + m_i = \frac{1}{2} m v^2 + \frac{1}{2} K x^2 =$$

$$m_i = \frac{1}{2} m^2$$

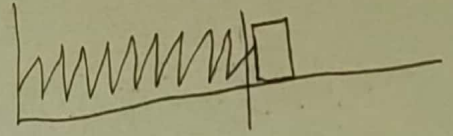
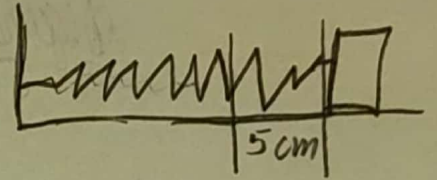
$$m_i = \frac{1}{2} m^2 \omega^2 A^2 = \frac{1}{2} (0.500)^2 (30)^2 (0.04)^2$$

$$m_i = 0.36 \text{ J}$$

$$4) \quad \frac{1}{2} Kx^2 = \frac{1}{2} mv^2$$

$$V = \sqrt{\frac{Kx^2}{m}} = \sqrt{\frac{500(0.05)^2}{2}}$$

$$V = 0.79 \text{ m/s}$$



$$5 \times 10^2 = 500 \text{ N/m}$$

$$5 \text{ cm} \times \frac{1 \text{ m}}{100 \text{ cm}} = 0.05 \text{ m}$$

$$5) \quad h = 500 \text{ Km} \quad m = 8.94 \times 10^{22} \text{ Kg} \quad r = 1815 \text{ Km}$$

$$g = \frac{GM}{R^2} = G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{Kg}^2$$

$$g = \frac{6.67 \times 10^{-11} (8.94 \times 10^{22})}{(1815 \times 10^3)^2}$$

$$h_{\max} = 500 \text{ Km}$$

$$V_f^2 = V_0^2 - 2gh$$

$$V_0 = \sqrt{2 (6.67 \times 10^{-11}) (8.94 \times 10^{22})} = 181254.4 \text{ m/s}$$

Tierra

$$h = \frac{V_0^2}{2g}$$

$$h = 1642694727 \text{ m} \div \frac{7 \text{ Km}}{7000 \text{ m}} = \underline{1642694.727 \text{ Km}}$$